Assignment – 07

Bank Loan Application Form (Advanced) — Advanced Fields Use Case

1. Objective

To extend the existing "Bank Loan Application Form" application in Zoho Creator by incorporating all seventeen Advanced Fields available in the Form Builder, so that the form moves beyond basic data capture and demonstrates automation, computed values, file/media handling, and structured lookups within a real banking use case.

2. Problem Statement

The basic Loan Application Form (built earlier using the thirteen Basic Fields) captures an applicant's identity, contact details, and loan preference, but it cannot handle several real-world requirements of a bank branch: verifying the servicing branch details against the master branch list, recording co-applicant details for joint loans in one submission, computing the estimated EMI automatically, issuing a unique scannable application tracking reference, or accepting supporting media such as income proof documents, a property walkthrough video for home loans, or a signed declaration.

Extend the Loan_Application_Form in Zoho Creator so that, in addition to the existing basic fields, it uses every Advanced Field type provided by Zoho Creator's Form Builder to support joint applications, EMI computation, branch verification, and automated application-ID/QR generation for tracking through the approval pipeline.

3. Scope of the Use Case

• Application Name: Loan Application & Pre-Screening Management (Advanced Edition) • Form Name: Loan_Application_Form (extended with Advanced Fields) • Primary Users: Prospective borrowers / Bank loan officers

- Report View: Loan_Application_Report (extended with computed EMI, application ID, and QR code columns)
- Objective of the Extension: Enable loan officers to verify branch assignment automatically, capture co-applicant details for joint loans, compute estimated EMI, and issue scannable digital application tracking references

4. Advanced Fields Used and Their Mapping

The table below maps every Advanced Field from the Zoho Creator Form Builder palette to its role in the extended Bank Loan Application Form.

Advanced Field	Form Field Name	Purpose	Sample Value / Behaviour
Decision Box	Has_Existing_Loan_With_Ba	decision on whether the applicant already holds another loan with the same bank	Yes / No toggle

Advanced Field	Form Field Name	Purpose	Sample Value / Behaviour

Rich Text	Loan_Terms_Display	Displays formatted loan terms and conditions (bold headings, bullet points, hyperlinks) directly within the form	"Interest rates are subject to change as per RBI guidelines. Processing fee is non refundable. Read the full terms below."
URL	Property_Listing_URL	Captures a link to the online property listing, for home loan applications	https://realestate.example.com/listing/44219
Image	Applicant_Photo	Captures a passport-size photo of the applicant for the bank's KYC record	applicant_photo.jpg
Percent	Interest_Rate_Offered	Captures the interest rate percentage offered to the applicant based on their credit profile	9.25%
Currency	Loan_Amount_Requested	Captures the loan amount the applicant wishes to borrow, formatted with currency symbol	₹15,00,000.00
Decimal	Debt_To_Income_Ratio	Captures the applicant's existing debt-to-income ratio, used in the eligibility assessment	0.32
Date-Time	Application_Submitted_On	Automatically records the exact date and time the loan application was submitted	24-Jul-2026 13:40
SubForm	Co_Applicant_Details	Captures repeating rows of co-applicant name, relationship, and monthly income for joint loan applications	Row 1: Meena Suresh – Spouse – ₹35,000; Row 2: Karthik Suresh – Son – ₹0

Lookup	Servicing_Branch_Lookup	Looks up and auto-fills the servicing branch name, IFSC code, and manager from a master Branches table	Anna Nagar Branch – IFSC BANK0001234 – Manager: R. Vasanth
Signature	Applicant_Digital_Signature	Captures the applicant's digital signature declaring that all information provided is true and accurate	Signed (drawn signature captured)

Advanced Field	Form Field Name	Purpose	Sample Value / Behaviour
File Upload	Income_Proof_Upload	Allows the applicant to upload salary slips, Form 16, or bank statements as income proof	income_proof_bundle.pdf

Audio	Loan_Purpose_Voice_Note	Allows the applicant to optionally record a short voice note elaborating on the purpose of the loan	voice_note_05.mp3 (0:25)
Video	Property_Video_Walkthrou	upload a short video walkthrough of the property being purchased, for home loan applications	property_walkthrough. mp4 (0:55)
Auto Number	Loan_Application_ID	Automatically generates a unique, sequential application reference number for every submission	BNK-LN-2026-000459
Formula	Estimated_Monthly_EMI	Automatically computes the estimated monthly installment using the loan amount, interest rate, and standard tenure	=(Loan_Amount_Request ed × Interest_Rate_Offered) / (12 × Standard_Tenure_Months)
QR/Barco de	Application_Track Automatica	a scannable QR code encoding the Loan Application ID for tracking through the approval pipeline	QR image encoding BNK-LN-2026-000459

5. Extended Form Layout (Section-wise)

The sections below are added to the existing Personal, Contact, and Income & Employment

sections of the basic Bank Loan Application Form.

Section E — Branch Verification & KYC

• Servicing Branch Lookup (Lookup field, auto-fills from master Branches table) • Applicant Photo (Image field)
• Income Proof Upload (File Upload field)
• Debt-to-Income Ratio (Decimal field)

Section F — Joint Application & Media Details

• Co-Applicant Details (SubForm field, repeating rows)
• Property Listing URL (URL field, for home loans)
• Property Video Walkthrough (Video field, optional)
• Loan Purpose Voice Note (Audio field, optional)

Section G — Loan Terms & EMI Computation

• Has Existing Loan With Bank (Decision Box)
• Loan Amount Requested (Currency field)
• Interest Rate Offered (Percent field)
• Estimated Monthly EMI (Formula field, read-only, auto-computed)

Section H — Declaration & System-Generated Values

• Loan Terms Display (Rich Text field, read-only reference block)
• Applicant Digital Signature (Signature field)
• Application Submitted On (Date-Time field, auto-captured)
• Loan Application ID (Auto Number field, system-generated)
• Application Tracking QR Code (QR/Barcode field, system-generated from Loan Application ID)

6. Validation & Configuration Rules

• Lookup field (Servicing Branch Lookup): connected to a master Branches form; loan officers type-ahead search and select rather than typing free text.
• SubForm (Co-Applicant Details): configured to allow a minimum of 0 and a maximum of 3 rows per application.
• File Upload (Income Proof Upload): restrict accepted formats to PDF, JPG, and PNG, with a maximum size of 5 MB.
• Image (Applicant Photo): restrict accepted formats to JPG and PNG only, with a maximum size of 2 MB.
• Video (Property Video Walkthrough): restrict maximum duration to 60 seconds and maximum size of 50 MB.
• Percent (Interest Rate Offered): restrict value range from 7 to 18.
• Currency (Loan Amount Requested): restrict to non-negative values only, formatted in INR (₹).
• Formula (Estimated Monthly EMI): set as a read-only field, recalculated automatically whenever Loan Amount Requested or Interest Rate Offered changes.
• Auto Number (Loan Application ID): configured with prefix "BNK-LN-", the current year, and a 6-digit auto-incrementing sequence.

- QR/Barcode (Application Tracking QR Code): configured to auto-generate immediately after the Loan Application ID is assigned on submission.
- Signature (Applicant Digital Signature) and Decision Box (Has Existing Loan With Bank) are mandatory fields; the form cannot be submitted without them.

7. Workflow Summary

- Step 1: Applicant completes the existing basic-field sections (personal, contact, and income/employment details).
- Step 2: Loan officer verifies the Servicing Branch using the Lookup field, which auto-fills the branch's IFSC code and manager.
- Step 3: For joint loans, co-applicant details are added as rows in the Co-Applicant Details SubForm.
- Step 4: Applicant uploads their photo, income proof, and optionally a property video walkthrough and a voice note on the loan purpose.
- Step 5: Applicant indicates any existing loan with the bank, and the Loan Amount Requested, Interest Rate Offered, and computed Estimated Monthly EMI are displayed. • Step 6: Applicant reviews the Loan Terms rich text block and provides their Digital Signature to confirm the accuracy declaration.
- Step 7: On Submit, Zoho Creator auto-generates the Loan Application ID, timestamps the Application Submitted On field, and generates the Application Tracking QR Code. • Step 8: The loan officer reviews the Loan_Application_Report, verifies uploaded documents and co-applicant income, evaluates the Estimated Monthly EMI against affordability norms, and uses the QR code to track the application through each approval stage.

8. Expected Learning Outcomes

- Understand the purpose and correct use case for each Advanced Field in Zoho Creator. • Learn how to configure Lookup fields for cross-form data verification. • Practice using SubForms to capture repeating, co-applicant data within a single submission. • Understand how Formula and Auto Number fields automate calculations and unique identifiers.
- Learn how to configure file-based fields (Image, File Upload, Audio, Video, Signature) with format and size restrictions.
- Understand how QR/Barcode fields can be used for real-world application-tracking validation and process automation.

9. Deliverable for This Lab Exercise

Students must extend their existing Loan_Application_Form in Zoho Creator to include all 17 Advanced Fields exactly as specified above, configure the listed validations and computed logic, and submit a screenshot-based lab record showing: (a) the extended form in Form Builder with all Advanced Fields, (b) a live preview of the form including the SubForm and computed Estimated Monthly EMI, (c) at least two sample submitted records showing the auto-generated Loan Application ID and QR Code in the report view, and (d) a screenshot of the Servicing Branch Lookup in action.